Identity and decision problem

Musanze Harvest Intelligence Service | Technical report

Identity field	Details
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Assignment	Musanze Harvest Intelligence Service

1. Stakeholder and practical problem

This project is designed for a potato cooperative in Musanze. Farm information can come from inconsistent spreadsheets, making it difficult for field officers to compare farms and make timely decisions. I developed a classroom AI prototype to estimate harvest quantity, classify production risk and group farms with similar conditions.

2. Questions addressed

Regression estimates potato harvest in tons from farm input conditions. Classification identifies whether the estimated yield per hectare is below the synthetic dataset median. Clustering groups farm records according to similar input conditions.

3. Scope and ethical boundary

I built a Python machine-learning pipeline, saved prediction models, a command-line interface and a FastAPI service. Because the training records are synthetic, this system is for learning and decision support only. It is not a verified forecast for real farmers and must not independently determine access to finance, insurance or public services.

4. Evidence basis

I used outputs from the reproducible pipeline, the saved evaluation reports and the recorded command-line and API tests. The report presents the final results after updating the training and model-selection workflow.

Data and reproducibility

Personal dataset generation and checks

1. Data generation

I used numpy.random.default_rng(7341). The seed is taken from the last four digits of my registration number. The assignment formula gives 240 + (7341 mod 41) = 242 original farm records. After data problems are introduced, the raw CSV contains 245 rows.

Feature	Generation rule
rainfall_mm	Uniform 600–1400
avg_temp_c	Uniform 12–26
soil_moisture_pct	Uniform 20–80
fertilizer_kg	Uniform 0–250
field_size_ha	Uniform 0.2–4.0
season	A or B
cooperative_support	0 or 1
yield_tons	Formula-generated target
high_risk	Below-median yield per hectare

2. Targets

Harvest yield is calculated from rainfall, moisture, fertilizer, field size, distance of temperature from 18°C, cooperative support, seasonal effects and normally distributed noise (standard deviation 0.80). The minimum yield is 0.50 tons. Risk labels use yield per hectare relative to its median.

3. Data-quality evidence

Measure	Observed result
Original / raw / cleaned rows	242 / 245 / 242
Missing soil moisture / fertilizer cells	12 / 13
Exact duplicates	3
Seed and code	7341 / AI-A2-7341

One appended duplicate can also duplicate a missing fertilizer value, explaining why the raw file contains 13 missing cells in that column. Repeated runs produced the same SHA-256 fingerprint:

408c124768bd6ff6bb83987a961dff1077a34ee08873b70363626f95b57e5fa5

Pipeline design

Preprocessing, model selection and leakage prevention

1. Training workflow

run_all.py checks the student identity and seed, generates the CSV, cleans duplicates, prepares the data, trains and evaluates three learning tasks, and saves model files and reports. Reusable logic is organized in the src folder.

2. Data separation

Both supervised tasks use the same stratified 80/20 training and testing split with random_state=7341. After cleaning, 49 records are reserved for final test evaluation. Models are chosen by cross-validation using only the training portion.

3. Preprocessing rules

Task	Implementation
Numeric imputation	SimpleImputer, median
Numeric scaling	StandardScaler where needed
Categorical imputation	Most frequent
Season encoding	OneHotEncoder, ignore unknown
Reusable process	Pipeline and ColumnTransformer

Within supervised training, the preprocessing pipeline is fitted to each training fold during cross-validation and then fitted on the training partition. Held-out testing uses those fitted transformations without recalculating them on test data.

4. Avoiding leakage

Only the seven permitted farm features are used as inputs. I exclude yield_tons, yield per hectare and high_risk from classification features. Regression models are selected using the lowest five-fold validation RMSE; classifiers are selected using the highest five-fold validation F1-score.

5. Saved models

The final pipelines are saved under models as regression.joblib, classification.joblib and clustering.joblib. The prediction tool and API load these saved pipelines instead of retraining.

Regression and classification

Model comparisons and held-out evaluation

1. Regression comparison

Model	CV RMSE	Test MAE	Test RMSE
DummyRegressor	2.5981	1.7906	2.2856
LinearRegression	0.9139	0.7048	0.8876
RandomForestRegressor	1.3827	1.0044	1.2344

I selected LinearRegression because it had the lowest mean validation RMSE (0.9139). On unseen synthetic test rows, its MAE was 0.7048 tons and RMSE was 0.8876 tons. A residual is the difference between predicted and actual harvest. Large residuals require inspection; For example, the test MAE of 0.7048 tons means the absolute prediction error averaged about 0.70 tons across 49 held-out records; it does not mean every farm had that error.

2. Classification comparison

Model	CV F1	Precision	Recall
DummyClassifier	0.0000	0.0000	0.0000
LogisticRegression	0.9276	0.9231	0.9600
RandomForestClassifier	0.9374	0.9259	1.0000

I selected RandomForestClassifier because it had the highest validation F1 (0.9374). On the shared test set, it achieved precision 0.9259 and recall 1.0000. This does not mean it will always identify every real high-risk farm.

3. Classification errors

Actual / predicted	Low risk	High risk
Actual low risk	22	2
Actual high risk	0	25

Two low-risk records were incorrectly flagged high-risk. No high-risk record was missed in this test. A missed genuinely high-risk farm would be important because staff might fail to check it. The sample is small and synthetic, so this outcome should not be generalized.

Clustering and interpretation

K-Means selection and farm profiles

1. Choosing K

Clusters (K)	Silhouette score
2	0.1576
3	0.1395
4	0.1347
5	0.1362
6	0.1414

I tested K-Means for K=2 through K=6 using scaled input features and a fixed random seed. K=2 had the highest silhouette score. However, a score of 0.1576 suggests weak separation, so these groups are only useful for exploratory descriptions.

2. Cluster profiles

Farm input	Cluster 0 mean	Cluster 1 mean
Rainfall (mm)	1012.733	1001.948
Temperature (°C)	18.870	18.494
Soil moisture (%)	51.035	49.715
Fertilizer (kg)	140.999	119.788
Field size (ha)	2.074	2.150
Cooperative support	0.0	1.0

The recorded numeric cluster summaries mainly differed in cooperative-support status and fertilizer use. Rainfall and temperature averages were close. These differences describe the generated records; they do not prove any cause of production differences.

3. Interpretation and caution

A field officer might use the groups to organize reviews of farms with similar input conditions. Cluster labels do not rank farms, and cluster 1 is not automatically better than cluster 0. The cluster grouping is descriptive and should not be used to judge farm quality or eligibility for support.

API and testing

Command-line predictions and service validation

1. Available interfaces

Endpoint or command	Purpose
GET /health	Report service health
GET /model-info	Show model names and verification code
POST /predict	Return harvest, risk, probability and cluster
predict.py --json	Predict using one local JSON record

The API accepts rainfall, temperature, soil moisture, fertilizer, field size, season and cooperative support. Typed validation is used to reject malformed or out-of-range requests.

2. Observed prediction

Output	Updated CLI result
Estimated harvest	14.710678798641739 tons
Predicted risk / probability	1 (high risk) / 0.805
Cluster	1
Verification	AI-A2-7341

3. Verification evidence

Test	Recorded outcome
pytest after refactor	8 passed, 1 warning
GET /health	HTTP 200
GET /model-info	HTTP 200
Valid POST /predict	HTTP 200
Invalid POST /predict	HTTP 422
Invalid CLI request	Missing-fields error and exit code 1

The only reported warning related to Starlette/httpx test-client deprecation; tests did not fail. Invalid rainfall of 2000 was also rejected during manual PowerShell testing before the refactor.

4. Commands

```
python run_all.py --student-id ug2527341@ines.ac.rw --seed 7341
python predict.py --json samples/valid.json
python -m pytest -q
uvicorn app:app --host 127.0.0.1 --port 8000
```

Model card and reflection

Limitations, responsible use and final review

1. Intended use

My project demonstrates a complete machine-learning service for classroom assessment. Its main functions are predicting harvest quantities, identifying relative production risk and grouping similar farms. It is not ready to support real agricultural decisions without further validation.

2. Limitations

The dataset was generated mathematically rather than collected from farmers. Real conditions include inaccurate measurements, changing weather, differences in soil and farming methods, and reporting errors. The risk label is based on relative yield per hectare, not a recorded crop failure. Clusters also have weak separation.

3. Human review and prohibited uses

The field officer or an agronomist should review surprising predictions and compare them with real observations. The service must not automatically determine access to loans, insurance, public services or agricultural support. If real farm data are collected, privacy and data security must be considered.

4. Suggested improvements

Future work would involve consented real farm records from different seasons, testing on unseen farms and years, monitoring model errors, examining differences across farm groups, and checking how reliable risk probabilities are. Access control and clearer version information would also be needed before deployment.

5. What I learned

One important lesson from this assignment is that I should not use the final test data to choose the best model. I improved the project by selecting models through training-only cross-validation and evaluating them with one shared test set. I also learned why saving pipelines, using a fixed seed and checking fingerprints matter for reproducibility.

6. Conclusion

The project demonstrates one reproducible workflow for generating farm data, training three types of machine learning, saving results and serving validated predictions. The best regression and classification models performed well on the generated test data, but the weak clustering separation and synthetic dataset mean that real-world performance remains unproven.